



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering
Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: I Semester B.E. CSE A

Course Code & Title: GE3151 Problem solving and python programming

Name of the Faculty member (s): Mrs. M.Swarna Sudha

Innovative Practice Description

Unit / Topic: Unit IV / Dictionaries: Operations and Methods

Course Outcome: CO 4

Topic Learning Outcome: TLO 11,12

Activity Chosen: Mind Map

Justification:

Dictionaries in Python have various operations and methods. A mind map enables students to organize this information in a hierarchical and structured manner, making it easier to understand the relationships between different elements. Drawing a mind map is an active learning activity that engages students in the process of synthesizing and summarizing information. It encourages them to think critically about the topic and actively participate in their own learning

- **Time Allotted for the Activity: 15 minutes**

Details of the Implementation:

- Instructor explained the particular concepts/topic in classroom within 35 minutes. Instructor gave a program to the students.
- Based on the discussion and the teacher asked the students to draw a mind map related to the topic within 15 minutes.
- Each student created a mind map on Dictionaries: operation and methods.
- Students represent the concept taught in the class by visual representation.
- DURGADEVI M: Focus on dictionary operations.
- MOHAMED AMMA GULSHAN M: Explore dictionary methods.
- Instructor collected the sheets from the students.

CO – PO / PSO mapping:

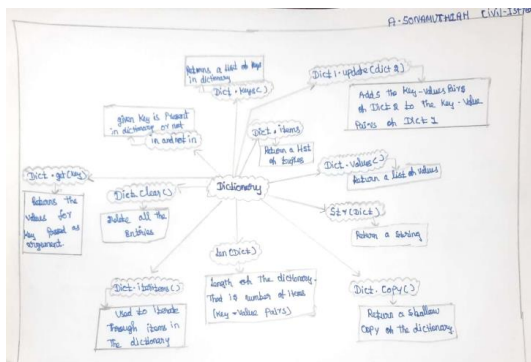
CO	PO1	PO2	PO3	PSO1
CO 4	2	1	1	1

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PSO1
	2	1	1	1
Justification for correlation	Students can able to use the compound data type dictionary for providing solution to the problems by using basics mathematics	Compound data types can be identified, formulated to solve the simple and complex engineering problems	Students can be able to provide solution to the real time decision making problems by using necessary compound data types.	Students can be able to apply compound data types in python program to provide the solution for information and communication technology related problems.

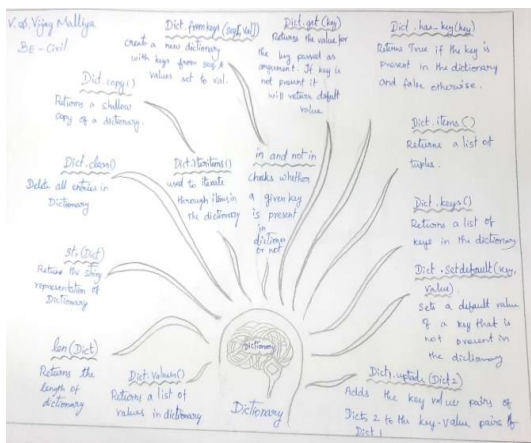
• Images / Screenshot of the practice:



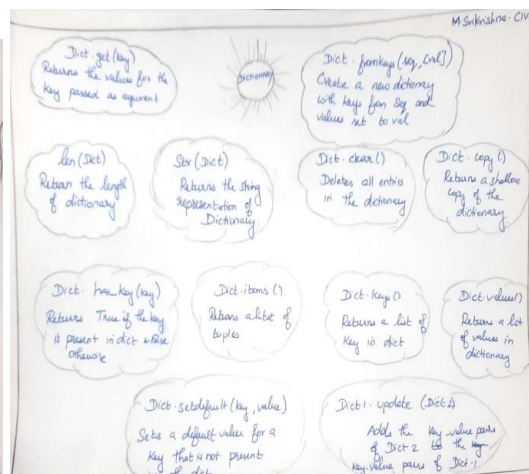
SRI PRISHIGAA R



SATHYAPRIYA K



SIVAPRIYA



A UDHAYASANKARI R

Fig:1 Glimpse of Student Mind Map

Reflective Critique:***Feedback of practice from students and other stakeholders:***

- Students expressed that they understood the concept very well and the pictorial representation made them to understand the topic clearly.
- This helped the students in recalling the topic taught on that specific day, generating new ideas about the topic.

Benefit of the practice:

- From this activity, the students can get more clarity in the particular topic.
- Through this activity, the students are able to remember and describe the various dictionary operations and its methods clearly

Challenges faced in implementation:

- Some of the students feel difficult to depict the concepts pictorially
- Some of them need some more clarification on the dictionary operations.
- Additional examples given to make them to understand the concept easily.

References:

<https://www.ritrjpm.ac.in/images/computer-science/Mind%20Map.pdf>
https://www.ritrjpm.ac.in/images/computer-science/43.CS6703_MindMap.pdf
https://www.ritrjpm.ac.in/images/computer-science/5_CS8591_Mindmap.pdf
<https://www.lucidchart.com/pages/how-to-make-a-mind-map>